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Real-time detection of insulator defects based on improved wavelet scattering convolutional network

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Abstract

To enhance the efficiency of drone inspection of transmission lines and promptly detect insulator defects for rapid repairs, we propose a hybrid model for insulator defect detection that can be deployed on drone edge devices, which can achieve online defect detection of high-definition inspection videos. The flexible light-weight models combined an improved wavelet scattering network with redefined modular coefficients, an optimized 4-layer convolutional neural network (CNN) model and 3-layer linear multilayer perceptron. When deployed on edge devices, their detection speed of 1080 * 720 images exceed 40 FPS, larger than the FPS of the inspection videos and is able to achieve real-time defect detection. The main reasons for the increase in speed are as follows: (1) The structural complexity of the improved wavelet scattering networks is much lower than that of CNNs. The extracted feature signals exhibit sparsity; except for the regions containing effective features, other regions have almost zero signals. (2) Instead of using ResNet with deep CNNs, a simple and optimized 4-layer CNN is employed with inputs either sparse features extracted by the wavelet scattering network or small-sized segmented images. Among our proposed models, improved wavelet scattering convolutional network (IWSCN) has an accuracy of 92.0% in flashover detection, which is close to you only look once (YOLO)-v7 but lower than the newer versions v8, v10, and v11. The CNN-4 model has higher accuracy in breakage detection than the v11 version and previous YOLO series algorithms by 4 percentage point. The proposed lightweight model represented by IWSCN has faster inference speed than YOLO series models and other edge device detection methods, achieving simultaneous optimization of defect detection accuracy and inference speed.

1. Introduction

The safe and stable operation of transmission lines is a necessary guarantee for the development of social production and people's livelihood [1]. Insulators provide isolation and mechanical support for power line conductors and play a crucial role in the operation of transmission lines. Due to long-term exposure to outdoor environments, insulators are prone to being affected by electrical, mechanical, and thermal stresses caused by different environmental conditions [2]. This can lead to defects of insulator such as contamination, flashover, and breakage, which will affect the stability of the power system. In severe cases, it may cause contact between power lines or between power lines and towers, resulting in large-scale power outages and thus significant economic losses [3]. Therefore, it is of great importance to promptly identify and replace faulty insulators [4].

Most transmission lines are erected in sparsely populated urban fringe areas or mountainous areas with complex terrain. The traditional manual inspection mode is time-consuming, laborious and difficult [5]. With the development of technologies such as unmanned aerial vehicle (UAV) and artificial intelligence (AI), as well as the construction of smart grids, more and more power grid units are using UAV to obtain inspection images of transmission lines and then using AI recognition algorithms to detect defects in

transmission lines. Now, an automated inspection pattern has been formed, with manual inspection as a supplement and UAV inspection as the main method [6].

Deploying the optimized AI insulator defect detection algorithm on edge computing devices of UAV to achieve local online (real-time) defect recognition has the following advantages: (1) the defect recognition is completed while capturing videos, making it unnecessary for the UAV to linger at the target, thus greatly decreasing the inspection time and enhancing the efficiency. (2) Recognition is completed locally, without the need to upload to the cloud, which can avoid the power consumption of high-definition image transmission and significantly increase the endurance time. (3) The maintenance personnel on site can be notified of the real-time defect detection results immediately and handle them right now, which greatly improves the efficiency of fault maintenance. (4) Localized image recognition solutions, which can easily and fully interact with smart grid supervisory control and data acquisition systems without manual intervention, help achieve fully automated inspection of UAV and greatly reduce labor costs.

However, due to the limited computing power of edge intelligent devices, existing mature AI detection algorithms may be difficult to run on them or run too slowly to achieve real-time detection. The processing speed of some detection algorithms based on traditional image recognition or machine learning is fast enough, but the resolution of their input images is insufficient, and the accuracy of object detection is relatively low, making it difficult to meet the reliability requirements of automated inspection of transmission lines.

To solve these issues, we have redefined the modulus coefficients of wavelet scattering networks and optimized the layers and structure of convolutional neural networks (CNN). We have flexibly combined the improved wavelet scattering network, a simple 4-layer CNN network, and a three-layer linear multilayer perceptron (MLP) to form a hybrid model for real-time detection of flashover or breakage defects in insulators. That is, the algorithm's inference time for high-quality images (resolution $>1080 * 720$) on edge devices is less than the interval time between two frames in UAV captured videos, while maintaining high recognition accuracy ($>90\%$). This solves the contradiction between improving detection accuracy and reducing inference time when deploying AI algorithms on edge devices. Our article's contributions mainly include the following:

- 1) By redefining and optimizing the modulus and scattering coefficients of wavelet scattering networks, the multi-level modulus or scattering coefficients can extract features of defects at different scales. Based on this feature extraction method, the improved wavelet scattering neural network (IWSNN) and the improved wavelet scattering convolutional network (IWSCN) can be used for rapid detection of flashover defects.
- 2) By analyzing the computational complexity of various types of layers in shallow CNN, it is pointed out that reducing the number of input image pixels and optimizing the layers can achieve a balance between inference time and accuracy in simplified 4-layer CNN (CNN-4) networks. Experimental verification shows that the ResNet structure improves the accuracy of shallow networks by less than 1 percentage point, but significantly increases inference time
- 3) A hybrid model represented by IWSCN and shallow CNN-4 has been proposed, which has better inference time than existing you only look once (YOLO) series algorithms and edge change detection algorithms. The inference speed for 720P on edge devices is higher than the video frame rate of 30FPS captured by UAV, and it can achieve real-time detection of insulator defects. Among them, IWSCN has an accuracy of 92.0% in flashover detection, which is close to YOLO-v7 but lower than the newer versions v8, v10, and v11. The CNN-4 model has better accuracy in breakage detection than the v11 version and previous YOLO series algorithms.

The remainder of this article is organized as follows. Section 2 reviews relevant research and recent advances on the topic. Section 3 discuss the framework and the principle of the method combined improved wavelet scattering networks and simplified 4-layer CNN. Section 4 presents the results and discussion of the combined model represented by IWSCN. Section 5 is the conclusion.

2. Related work

With the rapid development of AI algorithms, an increasing number of researchers have integrated them into their respective fields of study, achieving significant research results in areas such as image recognition, fault diagnosis, and pattern recognition. Currently, there has been extensive research on AI recognition algorithms for insulator defects in transmission lines.

2.1. Methods based on machine-learning

In the early days, researchers extracted features from images and utilized machine learning techniques to classify insulator defects, solving the time-consuming and labor-intensive problems of traditional manual inspection. Pernebayeva *et al* [7] used image normalization filters and machine learning classifiers to evaluate the status of outdoor glass insulators in harsh winter conditions and polluted environments. However, the accuracy of these algorithms is generally below 90%. Tiantian *et al* [8] extracted features such as Histogram of Oriented Gradients features and local binary pattern (LBP) features to train a support vector machine (SVM) classifier. The proposed method achieves a maximum accuracy of 83.5% with the fastest inference speed of 0.8 s. Wang and Zhang [9] transformed RGB images into the HIS space and used Gabor features based on SVM to distinguish insulators from the background. The proposed method achieves an accuracy of 75.2% with a peak inference speed of 6.12 s. Prasad and Rao [10] extracted features from insulators using wavelet transform and LBP histogram Fourier, and then used them to train an SVM classifier.

All the methods demonstrate relatively low detection accuracy less than 90%. Despite the low computational complexity, the traditional machine learning methods are difficult to deal with the target rotation and multiscale changes and easily affected by background noise and illumination conditions, which makes the recognition accuracy and robustness difficult to improve [11].

2.2. Methods based on CNNs

The emergence of CNN has successfully addressed the limitations of traditional methods. It can automatically learn and extract features from images, including textures, shapes, and patterns. Tao *et al* [12] developed a two-level CNN architecture for insulator localization and defect detection, necessitating intensive hyperparameter tuning and optimization at each stage. In response to the challenges associated with cascaded CNNs, researchers have explored image preprocessing techniques, such as wavelet scattering transform, image enhancement, and low-light enhancement. Lad *et al* [13] proposed an edge-oriented saliency object detection network that combines wavelet scattering network features with CNN-based features, enabling accurate identification of salient objects and effective detection of their edges. Cao *et al* [14] proposed an effective method for detecting glass insulator defects, focusing on insulator breakage defects. Yuan *et al* [15] proposed a novel algorithm that can significantly increase the contrast of low-light images, especially for object detection in overhead power transmission systems. This algorithm significantly improved the detection accuracy from 58.0% to 82.0%. Overall, image preprocessing techniques have greatly alleviated the limitations of CNN models by improving image quality, reducing noise, and enhancing generalization ability.

In recent years, CNN-based object detection algorithms have made significant progress in model architecture, which can be primarily categorized into two types: two-stage detection models and one-stage detection models. Two-stage detection models, with the R-CNN [16] series as a typical representative, mainly rely on region proposals for object detection. Wang *et al* [17] proposed a cross-domain multilevel feature alignment R-CNN network. It used a large number of artificially labeled images and a small number of unlabeled real images as the source and target domains, respectively. Based on Faster R-CNN, an instance-level feature adaptive alignment module was constructed. Different gradient adaptive training strategies was employed for the source and target domains to better achieve cross-domain instance-level feature alignment. In insulator defect detection experiments, it demonstrated high accuracy. Although two-stage detection methods exhibit the advantage of high accuracy in target localization, their detection speed is relatively slow due to the two-phase detection process. Single-stage detection models, such as single-shot detector (SSD) and YOLO, can directly detect objects without requiring separate region proposal networks and classifiers. Notably, the emergence of YOLO has revolutionized object detection by significantly improving both speed and efficiency. Yang *et al* [18] compared some of the deep learning models, YOLOv7, SSD, and DETR, for insulator detection based on small-sample learning. The in-domain and cross-domain scenario tests show that YOLOv7 is more suitable for insulator detection under small-sample conditions among the three models. Singh *et al* [19] conducted a comparative performance analysis of various models from YOLOv5 to YOLOv8, demonstrating the superiority of the YOLO series in transmission line defect detection. Cao *et al* [20] construct a lightweight model for insulator defect detection named coordinate attention mechanism (CAM) and feature channel shuffle operation (CSO) YOLO (CACS-YOLO). By introducing the CAM and CSO in the original YOLOv8m, they improved the detection precision and reduce the parameters of the model, compared with existing insulator defect detection models. The experimental results show that the mean average precision (mAP) of CACS-YOLO is 96.9%, and the parameters of the proposed model are 27.44 M. Wang *et al* [21] proposed the multiscale channel information (MCI)-global-local attention (GLA), a plug-in designed for YOLO series models, featuring two modules: the MCI extraction module and the GLA based on context information module (GLA-CI). MCI comprehensively extracts and utilizes multiscale feature map information, while GLA-CI captures both global context

information and local spatial details, thereby augmenting the learning capability of networks. Experimental results indicate that the MCI-GLA plug-in improves the AP of YOLOv4 to YOLOv8 models in detecting insulator breakage defects by 7.3%, 4.6%, 4.5%, 4.0%, and 5.3%, respectively. Zheng *et al* [22] proposed an Insulator Lack-YOLO (IL-YOLO) based defect detection model for insulators and enhanced its performance through model improvements. The experimental results demonstrate that IL-YOLO achieves a detection accuracy of 91.2%, representing a 3.6% improvement over the YOLOv5 algorithm, while maintaining a detection speed of 90 frames per second (FPS). Furthermore, it shows performance enhancements with a 0.5% increase in precision, 6.3% improvement in recall rate, and 3.8% boost in F1-score.

The CNN-based detection algorithm represented by the YOLO series can achieve high detection accuracy in various transmission line defects, and has a fast inference speed. It can usually perform real-time detection on 30FPS videos in mainstream graphics card computing devices. However, when deployed directly on edge devices with limited computing power without special customization and modification, the inference time is usually significantly reduced and positively correlated with the computing power of the device.

2.3. Detection method optimized for edge deployment

As the convolutional layers for object detection deepen, the number of parameters and the computational complexity of the network model keep rising. Consequently, the model size and computational cost also increase rapidly. As a result, the vast majority of object detection network models in deep learning are difficult to deploy on edge devices. To enable the deployment of object detection network models on embedded devices, some researchers utilize lightweight networks to better balance detection accuracy and inference time [23]. The synergy between UAV and embedded systems enables the automatic inspection of high-voltage insulators in challenging remote locations, reducing the need for manual inspection and enhancing the overall reliability of the distribution network. Waleed *et al* [24] devised a quadcopter-based system to classify ceramic insulators as healthy, broken, or dirty. However, limitations in detecting healthy insulators in onshore processing mode were observed, with mAPs of 0.67 onshore and 0.26 onboard. To address this, Rahman *et al* [25] proposed a solution using image augmentation, super-resolution, low-light enhancement, obstacle avoidance, UAV path planning, and YOLOv4 object detection. This approach achieved an *F1* score of 0.81 for detecting breakage insulators and a more efficient flight strategy. In addition, Song *et al* [26] introduced a collaborative intelligent method combining cloud and edge computing to detect insulator string defects. Although it only addressed two types of defects described in the CPLID dataset, it demonstrated good accuracy. Panigrahy and Karmakar [27] proposed a single-stage object detector method to address the limitations of traditional inspection methods by inspecting outdoor insulators using a hexacopter.

The above works have optimized the YOLO series models or CNN network, greatly improving their inference speed on edge devices. However, in general, its inference speed for images is usually seconds level, with the fastest inference time longer than 2 s, making it still difficult to achieve real-time detection of 30FPS videos.

3. Improved wavelet scattering network with CNN

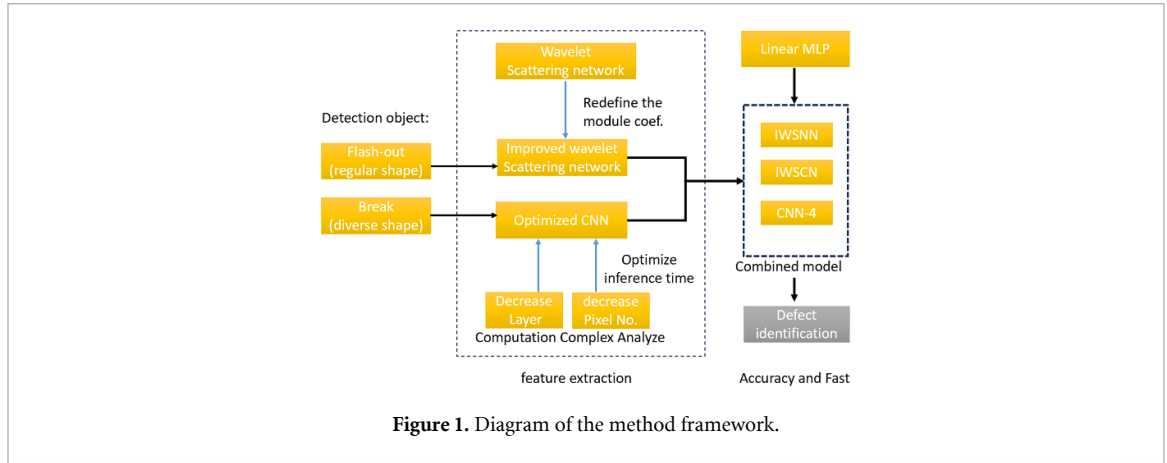
3.1. Frameworks of detection methods

Figure 1 shows the diagram of the proposed method and the research approach. The goal of our method is to perform insulation defect recognition with high accuracy and fast inference speed. The objects for defect detection can be categorized into two types. One type is the flashover on insulators, which is induced by lightning impulse and corona discharge and has a regular shape. The other type is breakage insulators caused by external force, whose shape is diverse and irregular, and some of which are easy to confuse with external objects. To make a compromise between the recognition accuracy and inference speed, two different models are utilized to extract the defect features.

For the regular flashover defect, we improve the wavelet scattering network by redefining its high-order modulus coefficients. The improved wavelet scattering network can greatly reduce the input information, effectively resolving the contradiction between accuracy and inference time on edge intelligent devices.

For the diverse break defects which are difficult to recognize, the performance of wavelet feature extraction is not ideal, and CNN is still needed to ensure the accuracy. Analysis of the computational complexity is conducted on the forward propagation to find the approach to decrease the inference time.

Based on the above improvement, we build combined insulator defect detection models, one combining the improved wavelet scattering network and MLP (IWSNN), one combining the improved wavelet scattering network and CNN (IWSCN), and the other is barely 4-layer CNN models (CNN-4).



3.2. Improved wavelet scattering network

The structure of wavelet scattering networks resembles that of CNNs, with the key difference being that their filters are predetermined wavelet filters. Notably, the training process does not require backpropagation, significantly reducing the influence of training samples on the model [28]. By computing semi-discrete wavelet transforms followed by nonlinear modulus operations, the resulting signal representations exhibit excellent properties such as translation invariance and deformation stability, precisely meeting the fundamental requirements for feature extractors in machine learning [29]. Moreover, these fundamental properties have been rigorously proven mathematically, addressing the theoretical limitations of CNNs [30, 31]. Extensive experiments have demonstrated that wavelet scattering networks achieve outstanding performance in handwriting recognition, texture classification, and audio classification tasks. Particularly in small-sample scenarios, they yield lower error rates compared to CNNs [32–36]. Consequently, wavelet scattering networks hold distinct advantages for recognition and classification tasks.

The number of network layers of the classical wavelet scattering network is determined by the number of wavelet bases in the preset filter, which limits the flexibility of the network structure. In this paper, the classical wavelet scattering network structure is improved so that the number of network layers can be unrestricted by the number of wavelet bases, and a small number of wavelet bases can be used for multilayer decomposition of high-resolution images.

The modulus coefficients and scattering coefficients of improved wavelet scattering network are defined as follows:

The 0th order scattering coefficient S_0 is expressed as:

$$S_0 x = x * \phi(u) \quad (1)$$

where S_0 is the 0th order scattering operation operator, x is the input 2D image information, $*$ is the convolution symbol, and $\phi(u)$ is the scale function of any of the preset wavelet bases.

The 1st order modulus coefficients and scattering coefficients are defined as:

$$U_1(u, \lambda_1) = |x * \psi_{\lambda_1}(t)| \quad (2)$$

$$S_1(u, \lambda_1) = U_1(u, \lambda_1) * \phi_{\lambda_1}(u) \quad (3)$$

where U_1 is the 1st order modulo operation operator, S_1 is the 1st order scattering operation operator, $\phi_{\lambda_1}(u)$, $\psi_{\lambda_1}(t)$ are the scale function and wavelet function of the wavelet basis, respectively, and λ_1 represents the label of the wavelet basis.

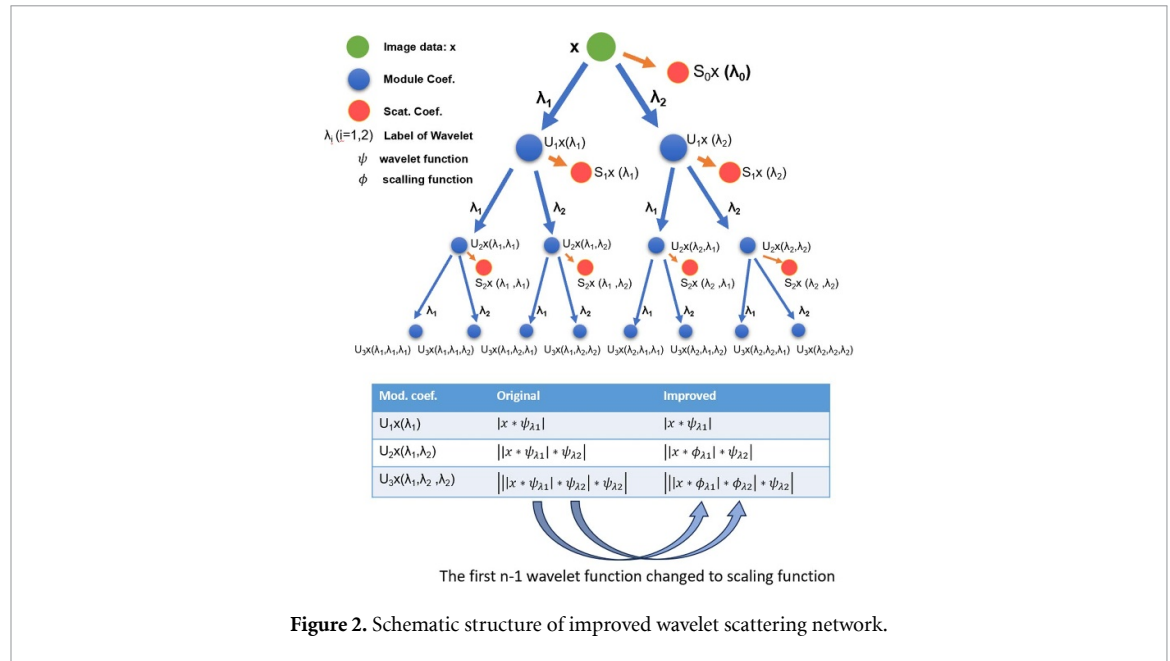
The 2nd layer of U_2 and S_2 can be expressed as:

$$U_2(u, \lambda_1, \lambda_2) = ||x * \phi_{\lambda_1}(u) * \psi_{\lambda_2}(t)| \quad (4)$$

$$S_2(u, \lambda_1, \lambda_2) = U_2(u, \lambda_1, \lambda_2) * \phi_{\lambda_1}(u). \quad (5)$$

And so on, the m th layer's modulus coefficient and scattering coefficient are:

$$U_m(u, \lambda_1, \dots, \lambda_m) = || \dots |x * \phi_{\lambda_1}(u)| \dots * \phi_{\lambda_{m-1}}(u) * \psi_{\lambda_m}(t) | \quad (6)$$



$$S_m(u, \lambda_1, \dots, \lambda_m) = U_m(u, \lambda_1, \dots, \lambda_m) * \phi_{\lambda_1}(u). \quad (7)$$

The improved wavelet scattering network redefines the high-order modulus coefficients. As shown in figure 2, compared to the original definition, the first n_1 wavelet functions ψ in the formula of the improved definition of n th-order coefficients have been replaced with the scaling function ϕ .

For the 2D image, the low frequency coefficients and the high frequency diagonal coefficients are selected as the convolution kernel of the improved wavelet scattering network, and the two expressions are:

$$\phi(x_1, x_2) = \sum_{n_1} \sum_{n_2} h(n_1) h(n_2) \phi(2x_1 - n_1, 2x_2 - n_2) \quad (8)$$

$$\psi(x_1, x_2) = \sum_{n_1} \sum_{n_2} g(n_1) g(n_2) \phi(2x_1 - n_1, 2x_2 - n_2) \quad (9)$$

where $\phi(x_1, x_2)$ is the scale function, $\psi(x_1, x_2)$ is the wavelet function, h is the low-pass filter, g is the high-pass filter, and n_1 and n_2 are the length and width of the input image.

The Daubechies (db) wavelets are widely used in fields such as signal processing, image processing, and data compression. Their orthogonality, compact support, and multi—resolution analysis capabilities endow db wavelets with unique advantages when dealing with complex signals and images. Therefore, in the wavelet scattering network of this paper, the wavelet basis is selected from the db series of wavelets. Considering the computational complexity of the model, the simplest db0 wavelet is ultimately chosen.

In the defect detection of transmission lines, it is of great significance to use the improved wavelet scattering network to extract key features. Due to the long-spanning and wide-spreading nature of transmission lines, they are located in diverse environments. Moreover, there is a wide variety of insulators on transmission lines, resulting in significant differences in the backgrounds of the captured insulator images. Additionally, when UAV is taking aerial photographs, the relative sizes of the insulator in the captured pictures may also change due to the different distances of the camera from the target object. To address these challenges, this paper utilizes the modulus coefficients and scattering coefficients of different orders in the improved wavelet scattering network for feature representation, so as to adapt to different types of defects and shooting distances. Since both the breakage and flashover defects of insulators belong to small target recognition, which requires higher order coefficients for feature representation, we adopt high-order (above the third order) scattering coefficients and modulus coefficients to represent the defect features.

Figure 3 shows the feature extraction results of the improved wavelet scattering network for breakage insulators and flashover insulators. The original image is shown on the left, the fourth order scattering coefficients S_4 in the middle, and the third order modulus coefficients U_3 on the right.

As can be seen from figure 3, the defect features of both breakage and flashover of insulators in the image can be represented by the scattering coefficients and modulus coefficients. The scattering coefficients and modulus coefficients of the flashover defects are relatively obvious. For the breakage defects, the scattering

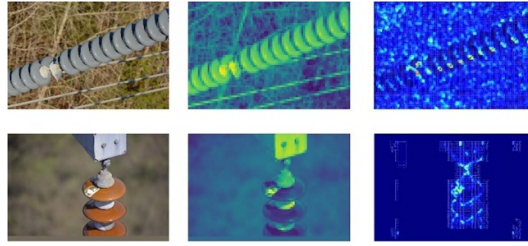


Figure 3. Effect of modulus coefficients and scattering coefficients.

Table 1. Comparison of recognition accuracy results for different channels of extracted defective images.

Y	R channel accuracy /%	G channel accuracy /%	B channel accuracy /%
Breakage	79.2	79.2	79.2
Flashover	90.3	90.3	90.3

coefficients are quite obvious, but the effect of the modulus coefficients is mediocre. Therefore, it is necessary to input the convolutional network module for further feature extraction, and then use the three fully connected (FC) layers module for recognition. It should be noted that the modulus images exhibit sparsity and most region has zero-value pixels, which is beneficial to decreasing the inferring time.

In order to reduce the amount of information in the input image, only one of the color channels of RGB image is used in this paper. In order to verify which color channel of R channel, G channel and B channel has the best effect, this paper conducts a comparison experiment. Firstly, the 3 color channels of the picture are extracted; then the corresponding modulus coefficient U and scattering coefficient S are obtained by inputting the improved wavelet scattering network respectively, in which the wavelet bases used in each layer of the scattering network are db0 wavelet; finally, the three FC layers module is used to identify and classify each channel. The specific experimental data are shown in table 1.

As can be seen from table 1, the recognition accuracies of the three channels are identical under the same conditions. We decide to use the B color channel in the insulator image as the input.

3.3. Improvement of CNN

1) Complex analyzes of network layers in CNN

As an AI algorithm, the CNN is widely used in the field of image processing due to its superior accuracy. However, limited by its dense computational and storage characteristics, the deployment of CNN at the edge has always been a popular research direction among scholars in the field of deep learning. Large models in CNN, such as DenseNet and YOLO, cannot be directly deployed at the edge due to their large number of parameters and high computational complexity. In order to achieve efficient deployment of large models at the edge, it is necessary to compress the model of the network structure and reduce the computational complexity. Currently, in terms of reducing the amount of computation and network storage, the model compression methods for CNN can be mainly divided into the following five types: parameter pruning, architecture search, network quantization, lightweight network design, and knowledge distillation [37]. After compressing large models, although the computational complexity and storage volume of the model can be reduced, edge intelligent devices still struggle to meet the computational complexity of a large number of convolutional operations, resulting in low computational efficiency. The structure of CNN usually consists of convolutional layers, pooling layers, non-linear activation function, and fully connected FC layers.

The computational complexity is the most commonly used metric to evaluate the size of a CNN model, also known as the number of floating-point operations (FLOPs). Assuming that the input image size is $N \times L$, the output size of the convolutional and pooling layers is represented as:

$$M_1 = \frac{N - D(K - 1) - 1 + 2P}{S} + 1 \quad (10)$$

$$M_2 = \frac{L - D(K - 1) - 1 + 2P}{S} + 1 \quad (11)$$

where N and L are the size of the input image's side length, K is the convolutional kernel size, P is the padding, and S is the stride and dilation factor D (generally defaulted to 1). Combining (10) and (11) can get

the formula for the computational complexity of the convolutional layer:

$$\text{FLOPs} = M_1 \times M_2 \times (K^2 \times C_{\text{in}} + 1) \times C_{\text{out}} \quad (12)$$

where C_{in} is the input channel and C_{out} is the output channel. Inside the parentheses is the sum of the multiplication and addition operands, the above equation considers the bias, if not, +1 will be neutralized.

When the pooling layer is maximum pooling, there is not a complete floating-point multiply and add computation in this layer, and together with the small amount of computation in the pooling layer, it can be considered as 0. When it is average pooling, its computational complexity is given by:

$$\text{FLOPs} = C_{\text{in}} \times W_{\text{in}} \times H_{\text{in}} \times K^2 \quad (13)$$

where W_{in} and H_{in} are the width and height of the input image and K is the pooling window size.

The FC layer multiplies the input tensor with the weight matrix and then adds a bias to produce the output as a linear transformation with the expression:

$$\mathbf{Y}_{i \times k} = \mathbf{X}_{i \times j} \mathbf{W}_{j \times k} + \mathbf{b} \quad (14)$$

where the input is $\mathbf{X}$, the output is $\mathbf{Y}$, $\mathbf{W}$ is the parameter to be learned by the model, $\mathbf{b}$ is the k -dimensional vector bias, i is the number of rows of the input vector, j is the number of input neurons, and k is the number of output neurons.

The FC layer is considered as a special convolutional layer, where the size of the convolutional kernel is equal to the size of the input image, and the formula for the computational complexity of the FC layer can be expressed as:

$$\text{FLOPs} = (I + 1) \times O \quad (15)$$

where I is the number of input neurons and O is the number of output neurons. The above equation is again considering bias and there is no +1 when it is not considered.

The activation layer usually employs the ReLU function, which introduces a nonlinear transformation into the neural network. The ReLU function is formulated as follows:

$$f(x) = \max(0, x) \quad (16)$$

where x is the input value. The computational complexity of the ReLU activation function is equal to the size of the input samples, and in the case of deep learning models, the activation function layer is negligible in comparison to the convolutional and FC layers.

The above texts analyze the computational complexity of various network layers contained in CNN networks. The area of the input image $N \times L$ directly affects the speed of the forward inference process. Therefore, reducing the area of the input image and appropriately segmenting the image can reduce inference time.

Another approach is to using the feature extracting results with sparsity by improved wavelet scattering networks. The calculated FLOPs using the equations (12), (13) and (15) suppose the image pixels has non-zero value. If the inputs exhibit sparsity and most region has zero-value pixels, the calculation of forward propagation would be simplified because the result of convolution or multiplication calculation with 0 can be directly judged as 0. Thus, using sparse image as input is equivalent to reducing the area of the input image.

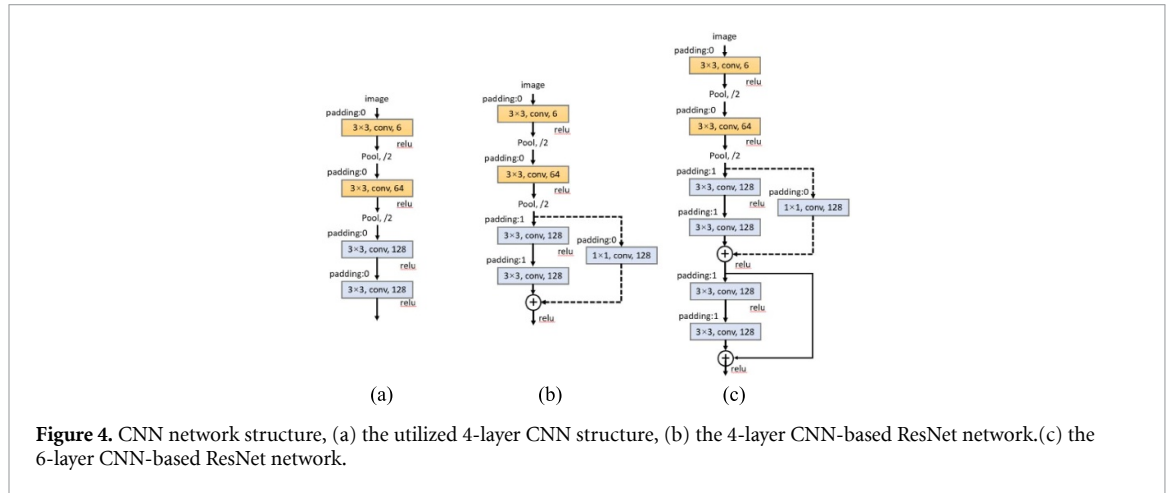
2) Number of layers optimize on CNN

At present, existing technologies can accurately locate and identify insulators and their defects on transmission lines to a certain extent. However, when handling real-time online detection tasks on edge devices, significant performance limitations still exist. The main reason lies in the fact that the huge computational complexity restricts the operational efficiency. Therefore, when designing the convolutional network module, the number of convolutional layers should neither be excessive, otherwise it will seriously affect the inference speed, nor be too few, otherwise the recognition accuracy cannot be guaranteed. To determine the number of convolutional layers, this paper has carried out a comparative experiment, using the IWSCN model with different convolutional layers to identify the breakage and flashover of insulator defects. The specific data are shown in table 2.

As can be seen from table 2, when the number of convolutional layers is 4, the IWSCN model has the highest recognition accuracy for both types of defects, reaching 86.9% and 92% respectively. Although the inference speed is the fastest (only 0.9 ms) when the number of convolutional layers is 0, the recognition accuracy for breakage defects drops to 79.2% in this case. Therefore, the final model in this paper adopts 4

Table 2. Comparison of recognition accuracy and speed across different convolutional layers.

Convolutional layers	Breakage accuracy /%	Flashover accuracy /%	Inference time (ms)
0	79.2	90.3	0.9
2	81.6	90.1	3.3
3	84.7	91.3	4.4
4	86.9	92.0	5.6
5	85.1	91.1	5.9

**Table 3.** Performance comparison of pure CNN and CNN-based ResNet architectures.

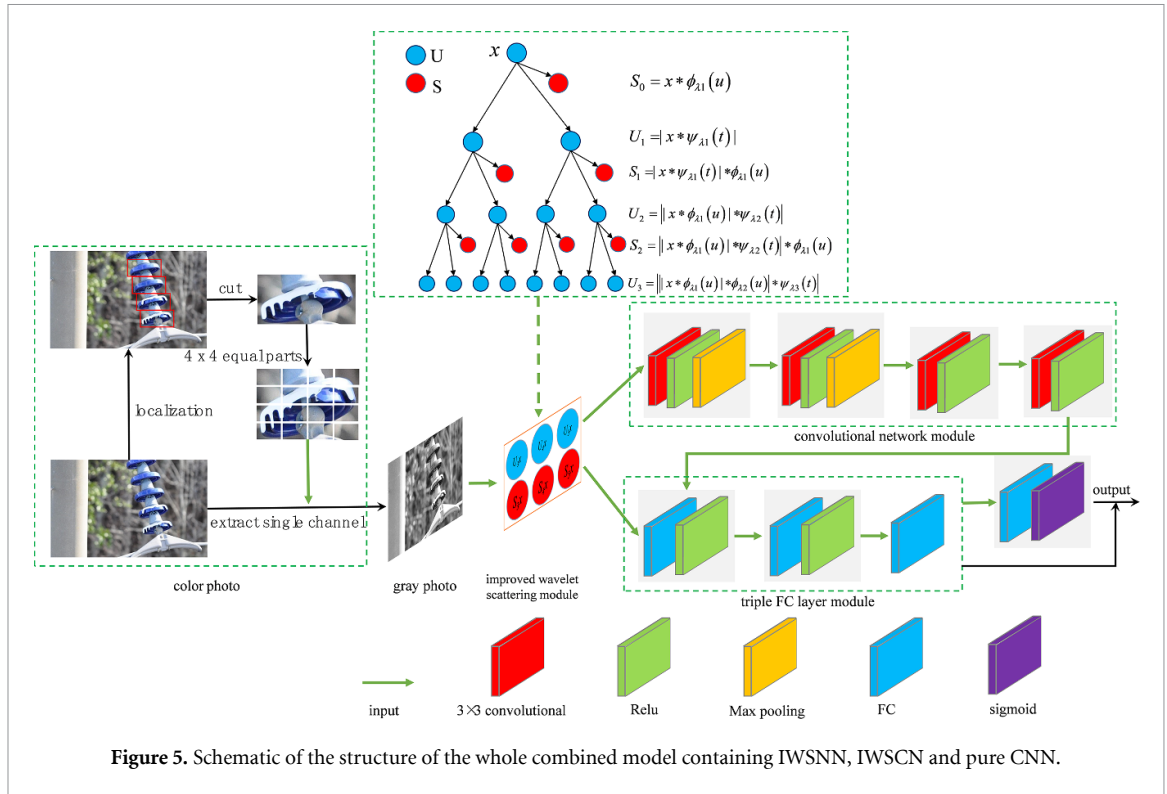
Convolutional layers	Breakage accuracy /%	Flashover accuracy /%	Inference time /ms
4-layer	86.9	92.0	5.6
4-layer-ResNet	87.5	91.7	6.1
6-layer-ResNet	88.2	92.4	9.3

convolutional layers. In CNN, Batch Normalization is an important component, whose main function is to standardize data so that it conforms to a certain distribution, thereby reducing internal covariate shift and help improve the training speed and stability of the model. However, we have proven through experiments that the effect of not adding Batch Normalization in the convolutional layer is better than adding it, and the best effect is achieved when the maximum pooling is added in the first two layers of convolution. So, the final four-layer convolutional network structure is shown in figure 4(a).

3) Influence of ResNet to CNN-4 model

The ResNet is designed to prevent gradient vanishing in deep networks, whose structures are connected the inputs and outputs of two or three CNN layers, and transfer the gradient of the input to the output. To investigate its influence on shallow CNNs, the comparison experiment is conducted between our basic CNN models and two simplified ResNet models. One ResNet model contains 4-layer CNNs and one Residual connection, while the other has 6 layers and two Residual connections. The three models' structures are illustrated in figure 4 and share nearly the same CNN layers, except that the padding of the last two/four layers changes from 0 to 1 in the two ResNet models. The Residual connection spanning layer 3 and 4 is built by a 1×1 CNN in order to transform the 64 channels into 128, and it is labeled by dashed lines in the figure. For Layers 5 and 6, their input and output have the same number of channels, and the residual is directly connected.

As can be seen from table 3, when used in the IWSCN, the three models in figure 4 have nearly the same accuracy in flashover detection, and their difference in accuracy is less than 1 percentage point. The two ResNet models' accuracy are higher accuracy than the bare CNN model in breakage detection, but are still under 90%. However, the inference time of the two ResNet models is increased by 8% and 66% due to the more complex structures. Considering both accuracy and inference time, the bare 4-layer CNN is selected in our combined model.



3.4. Proposed IWSCNs

Wavelet scattering networks can extract the boundary and texture features of images, but they are insufficient in the ability to fully utilize contextual semantic features in object detection tasks. CNN possess powerful feature extraction capabilities and the ability to combine multi-level semantics. Through multi-layer pooling and sub-sampling, CNN integrates multi-scale contextual data, but it still cannot separate the target data from the total data. Therefore, rationally combining these two networks can effectively improve the recognition accuracy.

The basic structure of the structure of the whole combined model is shown in figure 5. The improved wavelet scattering module first extracts a certain color channel of the RGB image to reduce the input information. Then, it takes this as the input of the wavelet scattering network. Finally, specific U and S are combined. The neural network module consists of two parts: the three FC layers module and the convolutional network module. If the combined image features are obvious enough, such as large-area flashover or breakage of insulators, only the simple three FC layers module needs to be input for recognition and classification. The improved wavelet scattering module is combined with it to form the IWSNN. If the defects in the insulators are small or blocked, the convolutional network module is required to further extract features, and then the three FC layers module is used for recognition and classification. The improved wavelet scattering module is combined with it to form the IWSCN. g power consumption. The bare 4-layer CNN model is called CNN-4.

4. Experimentation and analysis

4.1. Experimental condition and composition of data sets

Experiments are conducted on computer server with 64 bit Windows 10 and edge devices with Linux Ubuntu 20.04. The hardware configuration includes a 10-core Intel(R) Core (TM) i5-10500 CPU, a 12GB NVIDIA GeForce RTX 3060 GPU, and edge computing devices such as Jetson Nano, Raspberry Pi 4B, and so on. The programming language used is Python 3.10, and the deep learning framework chosen is Pytorch 7.1.3 and CUDA 12.2.68.

The IDID dataset [38] is adopted in this paper, which contains 1,596 aerial images of power inspections. These images display insulator samples with different defects, colors, sizes, and backgrounds, as specifically shown in figure 6. Among them, insulator breakage defects are marked with red bounding boxes, while flashover defects are outlined with yellow bounding boxes.

The experimental dataset is expanded to 2871 images through methods such as cropping, scaling, and rotation, ensuring that the image backgrounds of normal insulator data and defective insulator data are

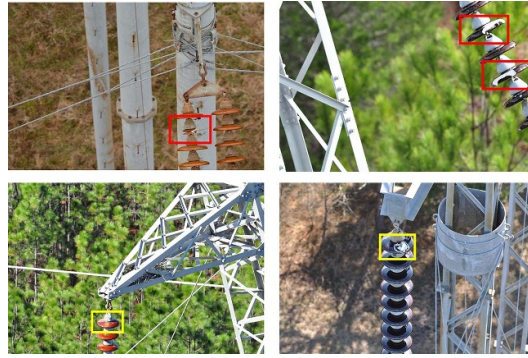


Figure 6. Defective insulator samples.

Table 4. Number and proportion of different image types.

Type of image	Number	Proportion /%
Flashover insulator	915	31.9
Breakage insulator	954	33.2
Normal insulator	1002	34.9

consistent. Among them, the proportions of flashover insulator, breakage insulator, and normal insulator images are 31.9%, 33.2%, and 34.9% respectively. The specific composition and proportions of the data are shown in table 4.

4.2. Evaluation metrics

The evaluation metrics for a model to detect images with insulator defects can be calculated through a confusion matrix, which generally encompasses the following four scenarios:

- 1) True positive (TP): when the actual image contains the target, and the detection result also indicates the presence of the target.
- 2) False positive (FP): when the actual image is a background or does not contain a target object, but the detection result incorrectly indicates the presence of the target.
- 3) False negative (FN): when the actual image contains the target, but the detection result incorrectly identifies it as a background or other type of objects.
- 4) True negative (TN): when the actual image is a background or contains another type of object, and the detection result also identifies it as not existing.

The evaluation metrics adopted in this paper include the following three:

Recall true positive rate (TPR): It represents the proportion of correctly detected samples containing insulator defects among all positive samples. The formula is:

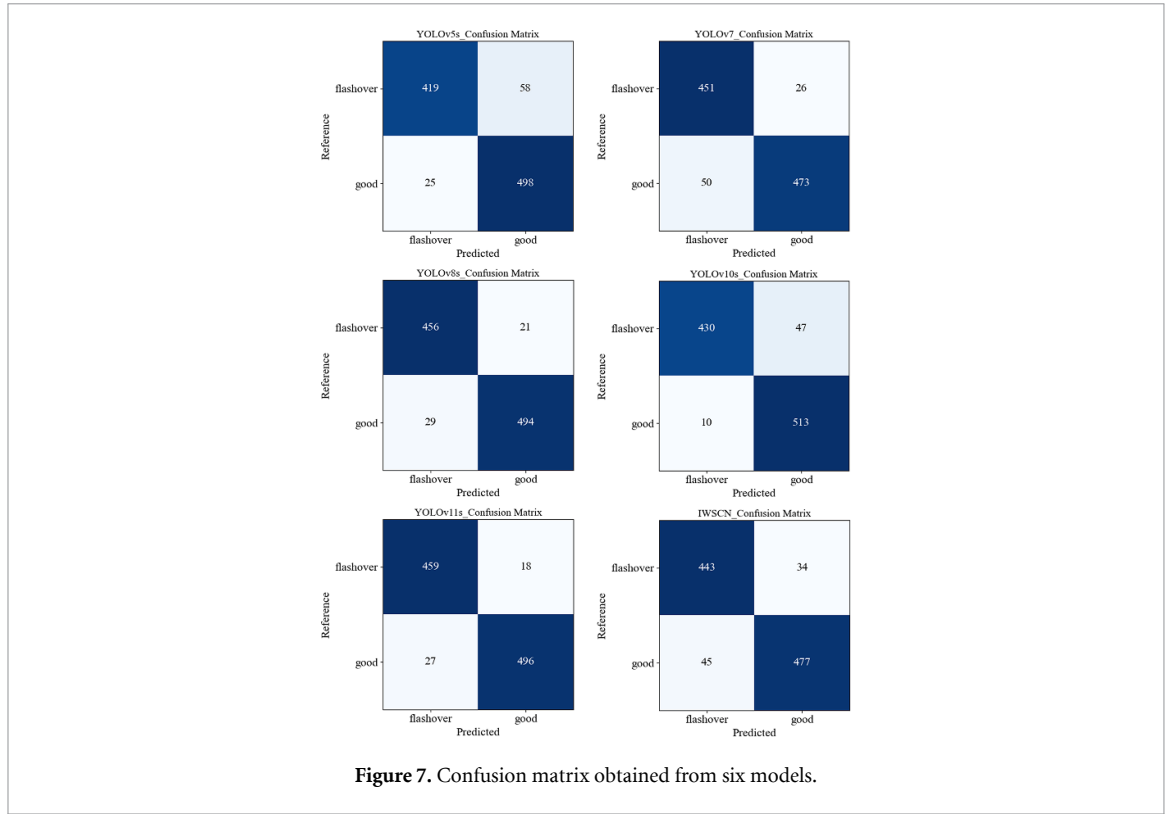
$$TPR = \frac{TP}{TP + FN} \cdot \quad (17)$$

Precision positive predictive value (PPV): The proportion of the number of samples actually containing insulator defects to the number of samples containing insulator defects determined by the classifier. The formula is:

$$PPV = \frac{TP}{TP + FP} \cdot \quad (18)$$

Accuracy (ACC): It represents the classifier's ability to judge the entire sample set, i.e. the proportion of correct detections to the total number of samples. The formula is:

$$ACC = \frac{TP + TN}{TP + FP + FN + TN} \cdot \quad (19)$$



mAP: It refers to the average of the Average Precision (AP) values across all categories, used to evaluate the overall performance of an object detection algorithm. The formula is:

$$AP = \int_0^1 PPV(TPR) dTPR \quad (20)$$

$$mAP = \frac{\sum_{i=1}^n AP(i)}{n} \quad (21)$$

where $AP(i)$ denotes the AP of the category, and n is the number of detected categories.

4.3. Analysis of recognition effects

The training condition of the IWSNN and IWSCN are described as follows: the learning rate is set to 0.001, the number of iterations is set to 200, a stochastic gradient descent algorithm is used to update the network weights with a momentum parameter of 0.9. The number of batches for each iteration is 150, and the number of batch samples is 200. In order to validate the performance of our proposed model, YOLOv5s, YOLOv7, YOLOv8s, YOLOv10s, and YOLOv11s models are introduced for comparative experiments.

1) Analysis of experimental results of flashover defects

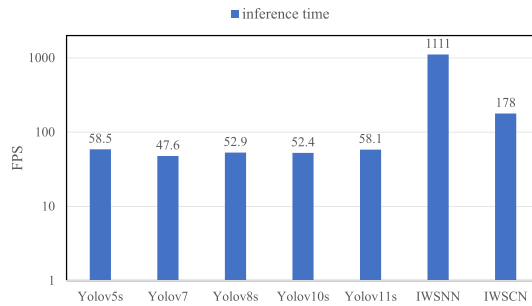
In this paper, flashover insulators and normal insulators were randomly divided into training and test sets. The training set consists of 438 flashover insulator images and 479 normal insulator images, while the test set contains 477 flashover insulator images and 523 normal insulator images. Seven models, namely YOLOv5s, YOLOv7s, YOLOv8s, YOLOv9s, YOLOv10s, IWSNN, and IWSCN, are used to recognize the 1000 sample images in the test set. All images were resized to 720P before being input into the network, and the respective confusion matrices were obtained. Figure 7 displays the confusion matrices of the five YOLO comparison models and the IWSCN model.

From the confusion matrices in figure 7, combined with (17), (18) and (19) can be used to find the three metrics for the six models. Along with the inference time of each model, their performance in detecting flashover insulators can be determined. The specific results are shown in table 5, which also includes the IWSNN model.

It can be seen from table 5 that Yolov11s performs outstandingly in terms of TPR and ACC, reaching 96.2% and 95.5% respectively. Yolov10s has the best performance in terms of PPV, reaching 97.7% while YOLOv5s had the fastest inference speed among the compared models at 17.1 ms. However, the models

Table 5. Comparison of the effectiveness of seven models in recognizing flashover defects.

Model	TPR/%	PPV/%	ACC/%	Inference time/ms
Yolov5s	87.8	94.4	91.7	17.1
Yolov7	94.5	90.0	92.4	21.0
Yolov8s	95.6	94.0	95.0	18.9
Yolov10s	90.1	97.7	94.3	19.1
Yolov11s	96.2	94.4	95.5	17.2
IWSNN	90.6	89.3	90.3	0.9
IWSCN	92.9	90.8	92.0	5.6

**Figure 8.** Comparison of inference speeds among the seven models.**Table 6.** Comparison of recognition performance for breakage defects across different models.

Model	ACC/%	Inference time (ms)	Para. Count/M	GFLOPs
Yolov5s	91	15.6	7.2	15.4
Yolov7	91.6	19.7	6.2	13.8
Yolov8s	94.7	17.3	11.2	28.8
Yolov10s	94.4	18.4	7.2	21.6
Yolov11s	95.6	16.0	9.4	21.7
IWSNN	79.2	0.63	66.3	6.6×10^{-3}
IWSCN	86.9	5.30	197.5	3.6
CNN-4	>99.75	10.2	311.9	6.7

proposed in this paper have significant advantages in terms of detection speed. The inference time of IWSNN is only 0.9 ms, which is 19 times that of Yolov5s. The inference time of IWSCN is only 5.6 ms, which is 3 times that of Yolov5s. Additionally, IWSCN achieved an accuracy of 92%, 0.3% higher than YOLOv5s, though slightly lower than the other compared models. The experimental results show that the model proposed in this paper has a clear advantage in terms of inference speed compared with other models. While ensuring high recognition accuracy, it also takes into account the lightweight and real-time performance of the network model. To more intuitively demonstrate the speed advantage of the models in this paper, the inference time of each model is converted into FPS and presented in a bar chart as shown in figure 8.

2) Analysis of experimental results on breakage defects

As can be seen from tables 1 and 2, the recognition accuracy of IWSNN for breakage insulators is only 79.2%, and that of IWSCN for breakage insulators is also merely 86.9%. Although the inference speeds of the two models are quite fast, their recognition accuracies do not reach 90%, making it difficult to apply them in practical scenarios.

To further improve the recognition accuracy of breakage insulators, the dataset was processed as follows: First, the insulator sheets in the images were located. Then, based on the location results, cutting was performed to obtain a single-piece insulator dataset. Subsequently, each single-piece insulator was divided into 4×4 equal parts, and the parts with defects and those without defects were separated to form two datasets, with a total of 6400 images. These data were then input into a CNN containing four layers of convolution (CNN-4) for training. After training, the trained model was used to test the newly divided 4×4 images of single-piece insulators. The test results were combined into 1×16 arrays to form a new dataset.

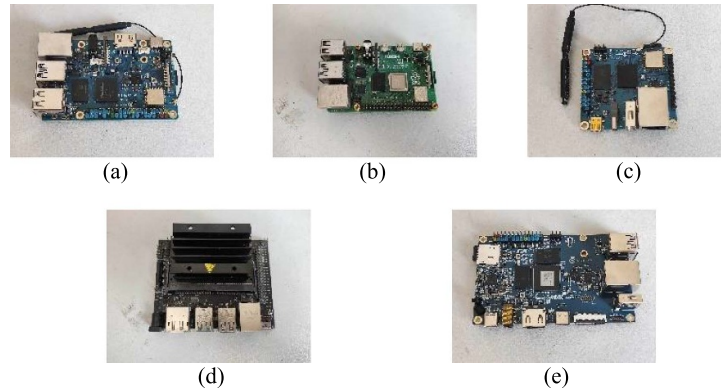


Figure 9. Physical drawings of the five edge devices, (a) Raspberry Pi 4B, (b) Orange Pi 3B, (c) Orange Pi Zero3, (d) Jetson Nano with 2GB graphic memories and (e) Orange Pi 5B.

The specific process is shown in figure 5. Finally, these arrays were input into the FC layer for training and testing at a ratio of 8:2. To verify the performance of the algorithm, YOLOv5s, YOLOv7, YOLOv8s, YOLOv10s, and YOLOv11s were also introduced for comparative experiments. This study randomly selected 551 individual insulator images for training and testing. The training set consists of 160 images of defective insulators and 140 images of normal insulators, while the test set contains 125 images of defective insulators and 126 images of normal insulators. The images were rescaled to 720p resolution when input into the network. The obtained experimental data is shown in table 6.

As can be seen from table 6, the recognition accuracy of the CNN-4 model for breakage insulators over 99.75%. Compared with IWSNN, it has increased by 20.5%, and compared with IWSCN, it has increased by 12.8%. The recognition accuracy has been significantly improved. It is 5.3% higher than Yolov8s and 4.1% higher than Yolov11s. In terms of inference speed, although the CNN-4 model has decreased to 10.2 ms, it is still nearly twice as fast as Yolov7 and Yolov10s.

The model's parameter counts and computational amounts, measured in GFLOPs, are also provided in table 6. The calculation of GFLOPs follows the conventions in the image recognition field, where 1 FLOP comprises one multiplication and one addition operation. This definition guarantees consistency with the formulas presented in this paper and aligns with the operation-counting libraries such as THOP, which are employed in YOLO series models. It can be observed that the parameter counts of YOLO series models typically hover around 10 million, with computational costs fluctuating around 20 GFLOPs. In contrast, the network parameter counts of our models are significantly larger than those of the YOLO series. For IWSNN, most of the parameters counts stem from the first FC layer, which transforms the input from a size of $3 * 960 * 320$ to 96 , and equal to 66.3 M. Since The deep network layers of IWSNN consist of only three FC layers, and the computational amount has the same value of parameters number. In the case of IWSCN and CNN-4, the parameter counts are 170.5 M and 311.9 M where FC layers account for 98.9% and 91%, while the computational amount are 3.6 GLOPs and 6.7 GLOPs, where CNN layers are the primary contributors and accounting for 95% and 96%. When compared to the top-performing YOLO11s, IWSNN, IWSCN, and CNN-4 exhibit FLOPs that are 0.3%, 16%, and 29% of YOLO11s' respectively, while their inference times are 3.9%, 33%, and 64% of YOLO11s' inference time. This indicates that there is potential for further decreasing our models' inference time by optimizing their algorithms and structures.

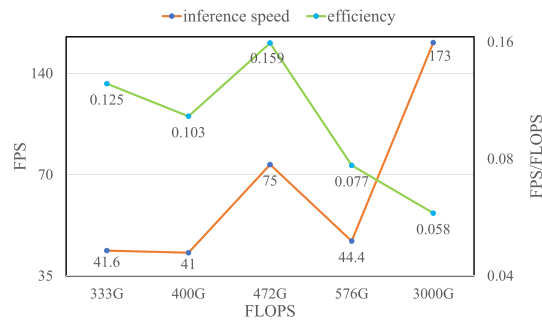
4.4. Analysis of inference speed of different edge devices

In order to verify the applicability and real-time performance of our algorithm, the IWSCN model is deployed to five different edge devices for experiments, including Raspberry Pi 4B produced by Raspberry Pi Foundation, Orange Pi 3B, Orange Pi Zero3 and Orange Pi 5B produced by Orange Pi, Inc. Jetson Nano produced by NVIDIA. The specific object is shown in figure 9.

The performance parameters of the five devices are shown in table 7, where the Orange Pi 5B has the highest peak arithmetic power of 3000 GFLOPs with ARM Mali-G610 MP4 CPU; the Orange Pi Zero3 has the smallest peak arithmetic power of 333 GFLOPs with ARM Mali G31 MP2 CPU; the Jetson Nano contains a 128-core Maxwell™ GPU, and the ARM processors of the other devices also contain high-performance image processing units. In addition, table 5 shows the detection speed of each edge device for 1280×720 resolution images, as well as the efficiency of their hardware optimizations. The hardware optimization efficiency is defined as:

Table 7. Five edge device parameters and their inference speed.

Edge device name	CPU/frequency/core	FLOPS	FPS	Efficiency
Orange Pi Zero3	Full-fledged H618 Cortex-A53/1.5 GHz/4	333 G	41.6	0.125
Orange Pi 3B	RMC RK3566/1.8 GHz/4	400 G	41	0.103
NVIDIA Jetson Nano	ARM Cortex-A57/1.43 GHz/4	472 G	75	0.159
Raspberry Pi 4B	ARM Cortex-A72/1.5 GHz/4	576 G	44.4	0.077
Orange Pi 5B	RMC RK3588S/2.4 GHz/8	3000 G	173	0.058

**Figure 10.** The relationship between peak computing power and detection speed.**Table 8.** Performance comparison of different algorithms.

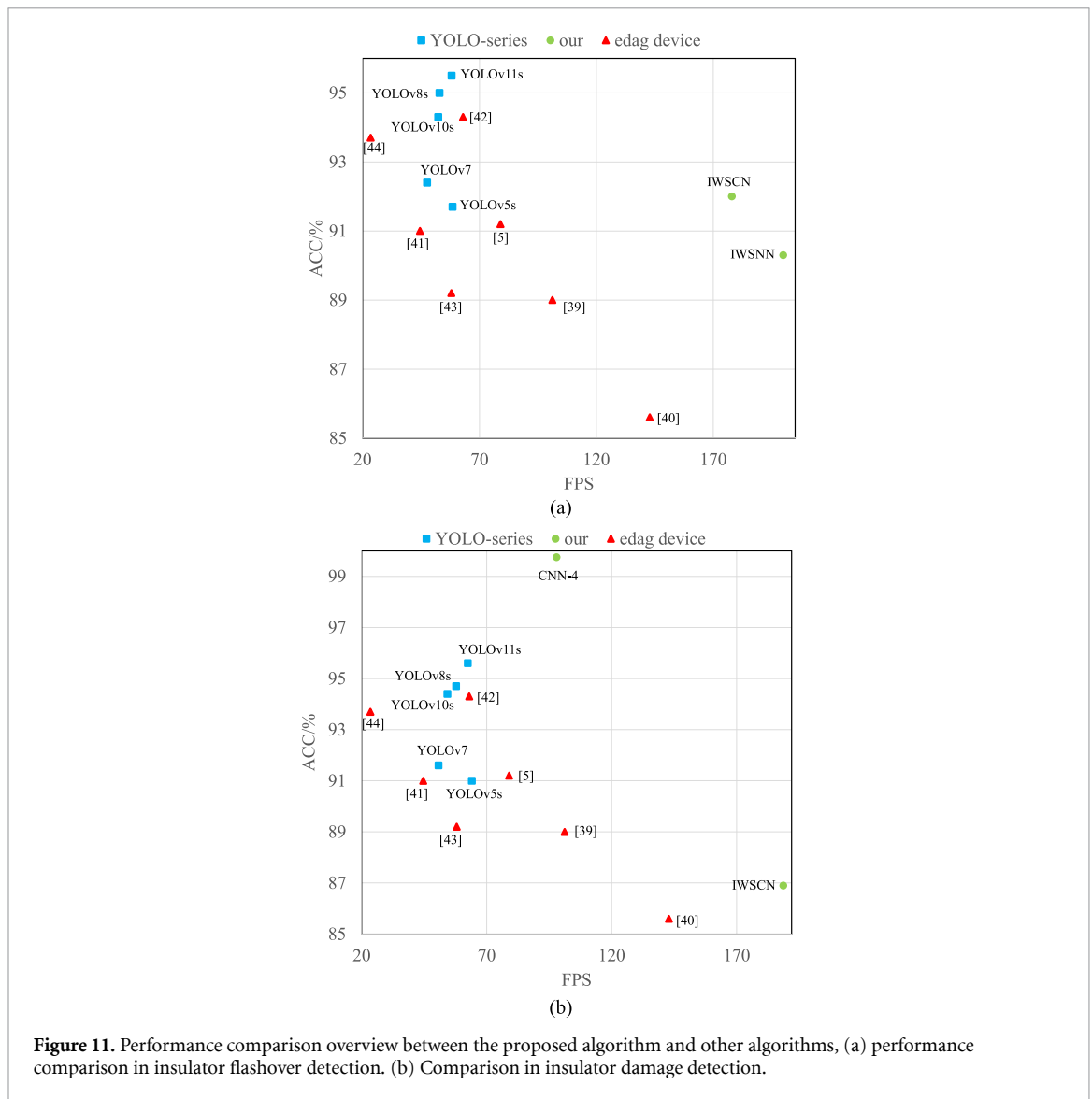
References	Image size	GPU detection speed/FPS	Edge device detection speed/FPS	mAP/%
[5]	640 × 640	53	79	91.2
[39]	300 × 300	101.2	—	89
[40]	640 × 640	277.8	142.9	85.6
[41]	640 × 640	—	44.6	91
[42]	640 × 640	111	63	94.3
[43]	640 × 640	—	58	89.2
[44]	640 × 640	285.7	23.5	93.5
Ours	1280 × 720	178	173	94.5

$$\text{efficiency} = \frac{\text{FPS}}{\text{FLOPS}} \quad (22)$$

where FPS is the detection speed and FLOPS is the peak arithmetic power of the device. Hardware optimization efficiency reacts to the matching degree and optimization efficiency between hardware and algorithm. The higher the matching degree, the higher the actual inference performance under the same arithmetic index.

As can be seen from table 8, our proposed model detects 1280 × 720 high definition (HD) images at a speed greater than 40 FPS on all five edge devices, with the fastest detection speed reaching 173 FPS on Orange Pi 5B, and 41 FPS on Orange Pi 3B, which has the relatively lowest performance and power consumption. In addition, the detection speed of the model shows a positive correlation with the CPU and GPU performance of the edge device, and the relationship between its peak arithmetic power and detection speed is shown in figure 10.

As seen in figure 10, the Jetson Nano has the highest degree of hardware optimization, which can reach 0.159 FPS/FLOPS, compared to other devices with similar arithmetic power, such as the Raspberry Pi 4B, Orange Pi 3B, and Orange Pi Zero3, its inference speed is nearly twice as fast as that of these three, which indicates that the Jetson Nano uses a better performing GPU. The Orange Pi 5B has the fastest detection speed and highest performance, but it has the least efficient hardware optimization at 0.058 FPS/FLOPS. The Orange Pi Zero3 has the lowest inference performance, but it has a level of hardware optimization for the algorithm that is second only to the Jetson Nano.



4.5. Performance comparison of other algorithms

To more intuitively demonstrate the superiority of the proposed method, we compare its performance with several state-of-the-art approaches proposed in recent years, as detailed in table 7. For fair comparison, mAP is employed as the evaluation metric for detection accuracy in the table 8.

As can be seen from table 8, although the input image size of our algorithm is 720P, which is a higher resolution than all other models, the detection speed on the edge device is the fastest among so models, reaching 173 FPS, which is over twice as fast as LEE-YOLOv7 model Hu *et al* [5], approximately three times faster than YOLO-GSS model Hou *et al* [43], and about four times faster than YOLOLS Wang *et al* [41]; Mobilenet + SSD Lu *et al* [39] can be deployed on edge devices, but their detection speeds are significantly lower than that of our algorithm; the reparameterized YOLOv5 Liu *et al* [40] and TD-YOLO Han *et al* [44] detect very fast on GPU, reaching 277.8 FPS and 285.7 FPS, respectively, which are about twice as fast as our algorithm, but their speeds decreases drastically when deployed to the edge devices, whereas our method suffers from a very small impact; Although the YOLO-inspection mAP Lu *et al* [42] is the highest among the references, at 94.3%, it is slightly lower than our algorithm's 94.5%, and its detection speed is lower than that of our algorithm both on the GPU and on edge devices. Therefore, our proposed algorithm ensures recognition accuracy while maintaining high real-time performance, enabling local online detection of HD video streams.

Figure 11 shows the performance comparison of IWSNN, IWSCN, and CNN-4 with YOLO series algorithms and other edge algorithms in insulator defect detection. Subfigure (a) compares the detection performance of flashover defects, while Subfigure (b) compares the detection performance of breakage defects. The horizontal and vertical axes represent inference speed and accuracy, respectively. The YOLO series models are represented by squares, the three models proposed in this article are represented by circles,

and the deployment algorithm of edge devices is represented by triangles. As shown in figure 11, IWSCN has an accuracy of 92.0% in flashover detection, which is close to YOLO-v7 but lower than the newer versions v8, v10, and v11. The CNN-4 model has better accuracy in damage detection than the v11 version and previous YOLO series algorithms. IWSCN and IWSNN are faster than all the other models, while CNN-4's detection speed for 720P images is only slower than [39, 40], which have input resolution 300 * 300 or 640 * 640.

5. Conclusion

To achieve on-site real-time detection of insulator defects in transmission lines with UAV, we propose a lightweight model based on an improved wavelet scattering CNN, which combines an improved wavelet scattering network with redefined modular coefficients and a small 4-layer CNN structure to extract defect features. When deployed on five edge devices, the detection speed of the model exceeds 40 FPS, and it can reach 173 FPS on Orange Pi 5B, which can realize local real-time detection of HD video streams while ensuring high accuracy.

The proposed combination models include ICWNN, ICWCN, and CNN-4. For flashover detection, the inference time of the IWSCN model is 5.6 ms, which is one-third of YOLOv5s (the fastest model in the YOLO series). Its accuracy is 92.0%, close to YOLO-v7, lower than v8, v10, and v11. For breakage detection, the recognition accuracy of the CNN model exceeds 99.75%, which is 4 percentage points higher than YOLOv11s. Its inference speed is 10.2 ms, which is faster than all the YOLO models.

In this article, only the db series wavelets were used to construct the scattering network, which resulted in insufficient accuracy for identifying breakage defects. In the future, we will try to combine multiple types of wavelets bases to improve the feature extraction ability of scattering networks for different types of defects. Introducing Transformer's attention mechanism into the scattering network may also help to further improve its accuracy.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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